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Power system expansion and renewable integration for drought-resilient electrification in Zambia

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HIGHLIGHTS

- Analysis of electricity access in Zambia using PyPSA-Earth and OnSSET.
- Solar and wind reduce hydropower dependency and mitigate climate risks.
- Investments in renewables and flexibility ensure supply security.
- Off-grid and hybrid solutions unlock cost-effective electricity access for millions.
- Policy and investments are critical for a resilient energy transition.

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ABSTRACT

Zambia's electricity sector relies heavily on hydropower, making it vulnerable to climate-induced hydrological variability, as seen in recent droughts causing severe load shedding and economic disruption. This study explores pathways to achieve universal electricity access by 2030 in Zambia, assessing the trade-offs between grid stability, climate targets, and equitable access. Using policy-driven scenarios, the analysis incorporates electricity demand growth, hydrological variability, and coal deployment to model system transitions. The results highlight that solar and wind energy are vital for diversifying Zambia's energy mix and reducing reliance on hydropower. Even in the absence of increased access efforts, at least 2.6 GW of solar power will be required by 2030, while under higher electricity access or low hydropower availability, solar could reach 8.4 GW and wind could exceed 4 GW. Flexible technologies such as biomass, coal, and battery storage are essential to maintaining grid reliability during dry hydropower periods. However, coal poses long-term sustainability concerns. This research adds to the emerging body of knowledge by incorporating off-grid solutions and demonstrating that achieving universal access requires a balanced approach. Strategies combining grid extension with decentralized solutions like solar home systems and mini-grids offer the lowest electrification costs per capita at $111 and $120, respectively, while achieving universal electricity access by 2030. In contrast, grid-dominant strategies are costlier and less inclusive. The paper concludes with policy recommendations, emphasizing the need to modernize infrastructure, support decentralized renewables, and mobilize investment, providing actionable insights for a resilient and inclusive energy transition in Zambia.

1. Introduction

1.1. Background

Achieving universal electricity access by 2030 is a key objective for Zambia's energy sector, aligning with the United Nations Sustainable Development Goal 7 (SDG 7) [1]. It is crucial for economic and social development, enhancing health, education, connectivity, productivity, and income levels. In sub-Saharan Africa, approximately 600 million people do not have access to electricity, posing a significant barrier to regional development. In response, the World Bank Group has committed to

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Nomenclature

List of Acronyms and Abbreviations

AfDB	African Development Bank
BAU	Business As Usual
BNY	Below Normal Year (Low Water Availability)
CAPEX	Capital Expenditure
CCG	Climate Compatible Growth
CGE	Computable General Equilibrium
GEP	Global Electrification Platform
GEGIS	Global-Energy GIS
GIS	Geographical Information Systems
HEAG	High Electricity Access Group
IEA	International Energy Agency
IMF	International Monetary Fund
IRENA	International Renewable Energy Agency
LCOE	Levelized Cost of Electricity
MV	Medium-Voltage
NC	No Coal Expansion
NY	Normal Year (Average Hydrological Conditions)
OPEX	Operating Expenditure
PPAs	Power Purchase Agreements
PV	Photovoltaic
REA	Rural Electrification Authority
REIPPPP	Renewable Energy Independent Power Producer Procurement Programme
REFiT	Renewable Energy Feed-in Tariff
RES	Renewable Energy Sources
RGC	Rural Growth Centers
SDG	Sustainable Development Goal
SEAG	Standard Electricity Access Group (Moderate Access)
tCO_2	Tonnes of Carbon Dioxide
WACC	Weighted Average Cost of Capital
ZESCO	Zambia Electricity Supply Corporation

providing electricity access to 300 million people across the region by 2030, underscoring the urgency of bridging this gap [2].

In Zambia, which has a population of approximately 20.8 million in 2024, only 53 % of people currently have access to electricity [3]. In addition, electricity access rates vary significantly across the country. While urban areas enjoy relatively high access, rural regions remain largely underserved [3]. This disparity necessitates targeted interventions to expand access to affordable and clean energy.

To address this challenge, the Zambian government, through the Rural Electrification Authority (REA), introduced the Rural Electrification Master Plan in 2008 [4]. It identified 1,217 un-electrified Rural Growth Centers (RGCs), which were grouped into 180 project packages aimed at accelerating rural electrification. Achieving this goal requires an estimated US$1.1 billion, with annual funding needs of approximately US$50 million from 2008 to 2030 [5]. Despite these efforts, Zambia's electrification progress has been slow. As of 2024, the rural electrification access rate stands at only 8.1 %, though progress has been made. By March 2024, the REA had electrified 545 RGCs, representing a 45 % RGC electrification rate.

The landscape of electricity production and installation costs has changed significantly in recent years. According to the World Energy Transitions Outlook [6] by the International Renewable Energy Agency (IRENA), the cost of producing 1 kWh of electricity with utility-scale solar PV has declined by 89 % between 2010 and 2022, making solar power installations more cost-effective. Furthermore, solar efficiency has improved from 14 % to 17 % over the same period. However, the costs of hydropower and geothermal energy installations have increased, posing challenges to Zambia's energy diversification strategy [7].

Zambia currently relies heavily on hydropower, which constitutes approximately 84 % of its electricity generation capacity [8]. This dependency makes the country vulnerable to climatic variations, particularly droughts, which have led to severe power shortages in recent years. For instance, the 2023–2024 droughts caused significant electricity shortages, forcing ZESCO to operate Kariba Dam at a restricted capacity of 215 MW, far below its nameplate capacity of 1,080 MW [9]. These power shortages have had wide-reaching economic impacts. Load-shedding increased to an average of 21 hours per day, severely affecting households, industries, and essential services [10]. Additionally, Zambia's economic outlook deteriorated, with the International Monetary Fund (IMF) revising its 2024 real GDP growth projection downward to 1.2 %, from an initial estimate of 2.3 %. Inflation surged to 15.7 % in October 2024 due to rising food prices and currency depreciation, exceeding the central bank's target range of 6–8 % [11].

Recognizing these challenges, the Zambian government has taken steps to diversify the energy mix. Initiatives such as the Renewable Energy Feed-in Tariff (REFiT) program aim to attract private investment in renewable energy projects, particularly solar photovoltaics (PV) and wind [8]. Despite these efforts, progress in diversification has been slow, highlighting the need for data-driven planning and modeling to identify resilient, low-carbon pathways for Zambia's energy future.

This study analyzes generation, transmission, and electrification pathways to achieve universal electricity access using the open-source tools PyPSA-Earth and OnSSET [12,13]. PyPSA-Earth will be used to determine the optimal energy mix required to meet electricity demand while considering climate challenges such as drought. It will also calculate the necessary transmission capacity, assuming all households are grid-connected, and allow for an electricity dispatch analysis under different scenarios. OnSSET will be used for assessing electrification strategies by supporting electrification planning and decision making for the achievement of energy access goals.

Grid electricity is typically generated in a centralized manner from large power plants, benefiting from economies of scale and lower generation costs. However, while electricity transmission and distribution costs remain relatively low in developed countries, expanding grid infrastructure requires substantial capital investment, long-term planning, and financial commitments [14]. A reasonable investment recovery period for such projects depends on a population with high purchasing power and significant electricity consumption [14]. In remote areas of developing countries like Zambia, extending the grid may not be economically viable due to low-income levels and minimal electricity consumption patterns, making off-grid electrification a cost-effective alternative [14].

To address this, OnSSET will assess grid and off-grid technologies and identify the most cost-effective technology for electrifying different population clusters in Zambia. This tool will provide insights into the number of required connections, the investment needed for both grid and off-grid electrification, the number of people who will benefit, and the capacity needed for each mini-grid technology. By integrating insights from PyPSA-Earth and OnSSET, this study will provide a comprehensive approach to achieving universal electricity access while ensuring reliability, affordability, and sustainability.

1.2. Literature review on electrification studies in Sub-Saharan Africa

In this section, we present a literature review of electrification studies in sub-Saharan Africa. Various studies have employed geospatial electrification modeling tools to analyze cost-effective pathways for expanding electricity access across the sub-Saharan region. Geographical information systems (GIS), often coupled with machine-learning techniques, hold great promise for informing policy on electrification planning. A number of studies have developed tools for least-cost strategies for electrification in rural areas [13,15–17].

The cost of electrifying households in 40 sub-Saharan African countries by 2030 has been analyzed in [18]. They used OnSSET to estimate levelized costs of electricity and per capita electricity costs. The findings highlight that solar-powered mini-grids and standalone systems can significantly reduce electrification costs in remote and high-cost areas, particularly for lower tiers of electricity access. Similarly, in Benin, a study conducted in collaboration with the World Bank developed an indicative least-cost geospatial electrification plan using OnSSET. This analysis examined 432 scenarios to determine the investment requirements for achieving universal electricity access by 2030. The results emphasized the importance of an integrated electrification strategy, combining grid connections with off-grid technologies to ensure nationwide access [19].

In Madagascar, the Royal Institute of Technology in Sweden participated in the Least-Cost Electricity Access Project alongside the World Bank and other stakeholders. They customized OnSSET to examine affordable electrification pathways, focusing on increasing access to electricity for households, enterprises, and health facilities. The project was structured into three components: (1) Grid electrification, which prioritized cost-effective grid extensions and densification in alignment with Madagascar's New Energy Policy and the national utility JIRAMA's extension plan; (2) Off-grid electrification, which promoted the deployment of standalone solar photovoltaic (PV) systems through financial institutions and off-grid solar companies to reach low-income households and remote areas; (3) Policy support for sustainable electrification [20].

In Somalia, a geospatial electrification model was developed to explore pathways toward universal electricity access under different timelines and levels of foresight. Using OnSSET, the study compared least-cost electrification options for 2030 and 2040. The results indicated that in the short-term, mini-grids and standalone PV technologies were the least-cost solutions under all but one scenario. However, over a longer timeline, constructing a national transmission backbone was found to be more cost-effective under conditions of high demand growth and lower centralized generation costs [21]. Another study in Somalia conducted a comprehensive techno-economic assessment of centralized and decentralized on- and off-grid systems for rural electrification [22]. The results demonstrated how integrated hybrid solutions with batteries can improve energy access while reducing costs and emissions.

Kenya has also been the focus of electrification modeling, where OnSSET and OSeMOSYS were soft-linked to capture both spatial and temporal dynamics in electricity planning. Two demand scenarios were considered: one representing low user consumption and another representing higher consumption levels. The findings suggest that geothermal, coal, hydro, and natural gas would form the optimal energy mix for the centralized national grid. In the low-demand scenario, standalone systems accounted for approximately 47 % of the electrified population. However, increasing electricity consumption shifted the optimal technology mix toward higher penetration of mini-grids and grid extensions [23].

In Zambia, a geospatial least-cost electrification study was conducted to evaluate on-grid and off-grid solutions for achieving universal electricity access by 2030 [24]. The country-wide study consists of two phases. First, they analyzed population density to identify sparsely populated areas suitable for standalone solutions. Second, they compared grid extensions and mini-grids using load forecasts and the levelized cost of electricity (LCOE) in the identified areas. The load forecast includes both population and agricultural electricity demands. Population demand is based on current customer consumption data, scaled by projected population growth and adjusted using historical consumption trends, while agricultural demand is estimated by electricity use for irrigation and post-processing. The findings provided a least-cost plan outlining medium-voltage (MV) grid extension feeders and designated areas for off-grid electrification.

These studies demonstrate the crucial role of geospatial electrification modeling in identifying least-cost solutions for expanding electricity access in sub-Saharan Africa. The findings highlight the need for integrated approaches that balance grid expansion with off-grid technologies to achieve universal access efficiently and sustainably.

1.3. Problem statement

Without intervention, Zambia's ambitions for sustainable economic development and universal energy access by 2030 may remain unattainable [4]. Despite its abundant natural resources and significant renewable energy potential, Zambia's over-reliance on hydropower creates substantial risks to energy security, economic resilience, and social well-being [8,25]. Frequent droughts, exacerbated by climate change, have demonstrated the system's fragility, necessitating urgent reforms to diversify energy sources and enhance system reliability [25].

1.4. Research questions

The guiding research question of this work is as follows:

What are cost-effective pathways for Zambia to ensure reliable and affordable electricity by 2030 under different electricity access levels and varying hydrological conditions?

This central research question is explored through the following subquestions:

1. How can Zambia reduce its dependency on hydropower while ensuring reliable and affordable electricity by 2030?
2. What are the trade-offs between achieving universal electricity access, integrating renewable energy sources, and maintaining grid stability under varying hydrological conditions?
3. What policies, technologies, and investment frameworks are most effective in enabling Zambia's transition to a resilient and sustainable power system?
4. What is the most cost-effective approach to integrating off-grid electrification solutions alongside grid expansion?
5. How can the combination of grid and off-grid solutions enhance electricity access while ensuring long-term sustainability and affordability?

1.5. Aim and key contributions

This study aims to model and evaluate resilient, low-carbon pathways for Zambia's power system by 2030. It seeks to expand equitable electricity access while mitigating hydropower dependency and optimizing renewable energy adoption in alignment with Sustainable Development Goals (SDGs) 7 (Affordable and Clean Energy) and 13 (Climate Action). This paper makes the following novel contributions to the literature on energy planning, electrification, and climate resilience in sub-Saharan Africa:

1. Development of a Dual-Model Framework for Integrated Electrification Planning: We present a unique combination of two complementary open-source modelling tools PyPSA-Earth and the OnSSET. The tools are used to evaluate both centralized grid expansion and decentralized off-grid solutions. This dual approach enables a comprehensive assessment of least-cost electrification pathways, a gap often overlooked in previous studies that focus on either grid or off-grid strategies in isolation.
2. Scenario-Based Analysis of Zambia's Energy Transition Under Climate Stressors: We construct and analyze twelve integrated scenarios that vary across electricity demand growth, hydrological variability, and coal expansion. This allows us to explore trade-offs between energy security, climate resilience, and universal access, providing a more nuanced understanding than static or single-variable models.

3. Validation and Customization of a High-Resolution Power System Model for Zambia: Building on the PyPSA-Earth framework, we develop a validated, country-specific model for Zambia, incorporating local data on hydropower inflows, generation capacity, and demand projections. This enhances the model's accuracy and policy relevance and contributes to the open-source modelling ecosystem.
4. Quantitative Assessment of Cost, Emissions, and System Resilience: We provide detailed estimates of total system costs (CAPEX and OPEX), CO_2 emissions, and load-shedding risks across scenarios. These metrics are contextualized with dispatch analyzes and generator duration curves, offering insights into the operational dynamics of a future Zambian power system under hydrological stress.
5. Policy-Relevant Insights for Low-Carbon Development and Investment Planning: The study offers actionable recommendations for policymakers, including the role of flexible technologies (biomass, batteries), the importance of diversified energy portfolios, and the need for innovative financing mechanisms to attract private investment. These insights are grounded in scenario results and aligned with Zambia's national development plans.
6. Contribution to the Literature on Electrification in Developing Countries: Unlike prior studies that focus primarily on technical feasibility or cost minimization, this work integrates socio-political and environmental dimensions such as climate vulnerability, rural access disparities, and institutional readiness into the modelling framework. As such, the findings in this study carry important implications for other developing countries facing similar challenges in electricity access.

The structure of the rest of the paper is as follows. Section 2 outlines the methodology, detailing the use of PyPSA-Earth for grid-based scenario analysis and OnSSET for assessing electrification strategies. Section 3 describes the development and validation of the PyPSA-Earth model tailored to Zambia's power system. Section 4 presents the energy scenarios explored in the study, focusing on generation, transmission, and electrification pathways. Section 5 discusses the results and analysis, highlighting key findings related to system performance and electrification outcomes. Section 6 examines the policy implications and summarizes key insights of the study. Section 7 reflects on the main findings and proposes directions for future research.

2. Methodology

In this section, an overview of the PyPSA-Earth methodology is provided, followed by a brief discussion of the OnSSET methodology, which is the underlying model behind the Global Electrification Platform (GEP). The methodology outlines clear steps that demonstrate how the two modeling tools are used to provide insights into universal electricity access in Zambia.

2.1. Pypsa-Earth

PyPSA-Earth is an open-source global energy system model designed to support high-resolution, large-scale energy system analysis with high regional and temporal resolutions [12] using the well-known PyPSA Framework [26]. PyPSA-Earth offers a comprehensive set of features tailored for large-scale energy system modeling. It supports flexible spatial scopes, ranging from global to detailed analyzes of specific subregions, and enables high temporal and spatial resolution for capturing dynamic system behavior.

PyPSA-Earth enables comprehensive integration of renewable energy sources, transmission networks, and sector coupling, making it particularly well-suited for evaluating decarbonization pathways and infrastructure planning. At the core of PyPSA-Earth is a reproducible workflow, which decomposes the modeling process into transparent and traceable steps. This structure enhances adaptability and ensures the scientific rigor required for large-scale energy system research. Its open-source modular structure, compatibility with publicly available global datasets, and community support ensure transparency and reproducibility, aligning with best practices in open-source energy system research [27].

Established frameworks such as PyPSA and its global extension PyPSA-Earth have been successfully applied to energy system modelling across various regions around the world, including developed regions such as the United States [28], China [29], Germany [30], the United Kingdom [31], and Spain [32], as well as in developing countries such as Bangladesh [33], South Africa [34], Nigeria [12] and Kazakhstan [35]. These applications demonstrate the framework's flexibility, scalability, and robustness in diverse policy, technical, and data environments. In light of these capabilities, PyPSA-Earth is selected as the modelling platform for this study to rigorously evaluate cost-optimal and technically feasible pathways for energy system transformation in Zambia.

The model forms a linear optimal power flow problem which aims to optimize generation, grid, and storage capacities by minimizing the annualized system costs using a discount rate of 7 % [36]. A detailed mathematical description is provided in the Supplementary Material and the detailed workflow of PyPSA-Earth is shown in Fig. 1.

The first stage involves gathering data from global repositories. Infrastructure data, such as power plant and electricity network information, is derived from sources including OpenStreetMap [37] and various power plant databases [6,38–40]. Environmental data, such as solar irradiation, wind speeds, and temperature, is obtained from datasets like ERA5 reanalysis [41] and SARAH-2 satellite data [42]. Socio-economic data, which includes population and GDP figures, comes from sources such as Murakami and Yamagata [43]. Additionally, non-geographic data, such as technology costs and emission targets, can be specified by users [12].

The second stage processes the data to fit the PyPSA network's spatial and temporal resolution [26]. This involves aggregation, estimating unknown values, and merging datasets. The third stage formulates a linear optimization problem aimed at minimizing annualized capital and operational costs while adhering to constraints such as demand satisfaction, transmission limits, renewable availability, geographical potentials, and emission targets. Open-source solvers like CBC or commercial alternatives such as Gurobi can be used to solve this problem. Finally, PyPSA-Earth offers tools for result analysis through Jupyter notebooks, providing examples and guidance for users unfamiliar with the ecosystem [44].

2.2. OnSSET: open-source spatial electrification tool

The need for open-source platforms for electrification is identified as a key pillar to achieve tangible progress in universal access to clean sustainable energy [45]. OnSSET is a GIS based bottom-up open-source optimization tool that has been developed to support electrification planning and decision making for the achievement of energy access goals [13].

Considering factors like population distribution, existing infrastructure, and different energy technologies, with different geospatial characteristics and socio- and techno-economic data, OnSSET identifies the least-cost supply option and helps policymakers make informed decisions about providing universal access to electricity [46]. OnSSET has been extensively used in literature, informing agencies such as the International Energy Agency (IEA) [47], the UN [48], The World Bank [49], as well as several developing countries such as Ethiopia [50], Nigeria [51], Burkina Faso [52] and many other countries in sub-Saharan Africa [18]. Moreover, a comparative study between the most widely used GIS-based rural electrification planning models concluded that OnSSET has better capabilities than the other models [53]. Another study reviewed 27 energy models and identified OnSSET as the only open-source tool suitable for analyzing electrification

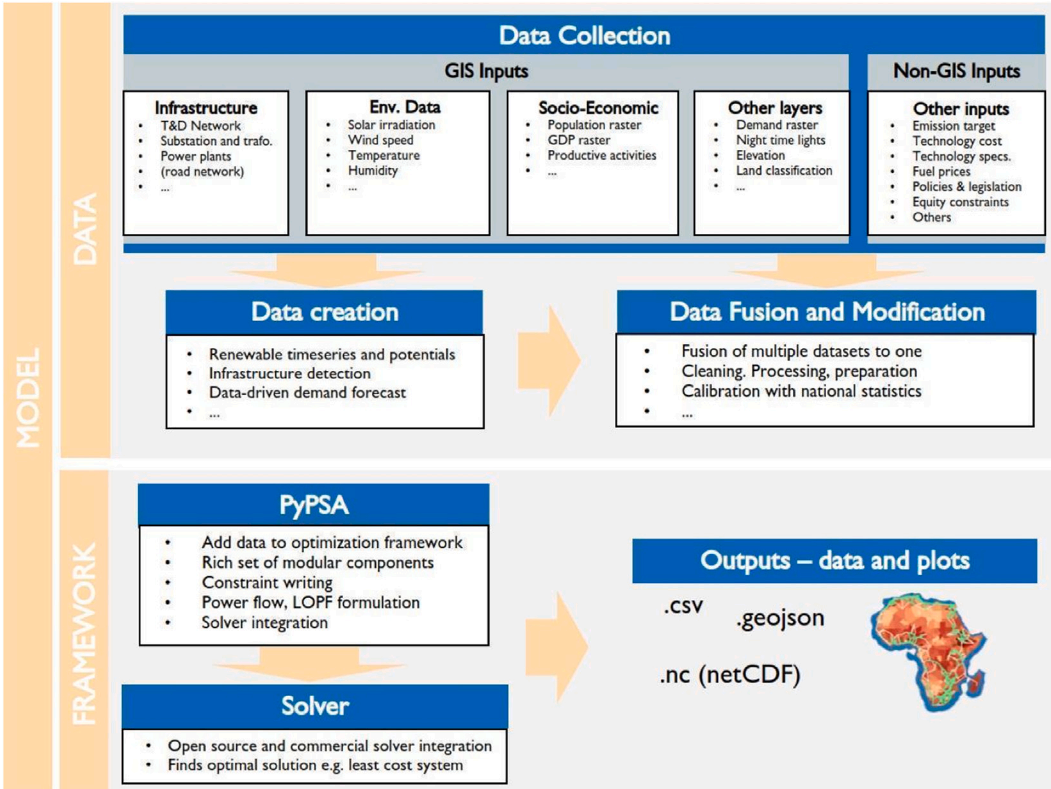


Fig. 1. PyPSA-Earth complete workflow. Source: [12].

in rural developing areas [15]. Given these capabilities, this study explores the off-grid space for electricity access in Zambia using OnSSET.

OnSSEt has an electrification algorithm that follows two complementary processes: GIS analysis and cost-based electrification modeling. The GIS analysis uses a range of indicators based on open-source geospatial data to determine the initial electrification status. These include proximity to the roads, proximity to existing transmission infrastructure, nightlights and population density. Using this information, the model then calculates the Levelized Cost of Electricity (LCOE) for several electrification technologies, considering local demand, network requirements, and resource availability. For grid extension, the algorithm connects non-electrified settlements only if the grid-based LCOE is lower than off-grid options and the medium-voltage (MV) grid extension does not exceed 50 km. Newly connected settlements contribute to future grid strengthening costs. If a settlement remains unconnected, the algorithm selects the least-cost mini-grid or standalone system to meet its electricity needs.

This iterative process continues until all feasible grid extensions are completed [46]. Fig. 2 below summarizes the methodology.

The Global Electrification Platform is an interactive tool, built on OnSSET, that facilitates the analysis of electrification investment scenarios across different countries. It evaluates pathways for achieving universal electricity access through grid expansion, mini-grids, and stand-alone systems within a defined timeframe (e.g., 2030). The model identifies the least-cost investment requirements based on publicly available data on demand and existing infrastructure [54]. The platform promotes transparency by allowing expert feedback and collaboration. The Global Electrification Platform consists of two main components: The Generator, which enables users to customize electrification scenarios by selecting parameters and assumptions, offering six main levers on the web platform and over 150 parameters in the full model; and The Toolbox, which provides tools for generating input data, allowing users to integrate and analyze their own GIS and non-GIS datasets for enhanced customization.

2.3. Research methodology

The methodology consists of two main parts. The first part involves analyzing scenarios using PyPSA-Earth, under the assumption that all loads will be connected to the grid. This analysis aims to determine the optimal generation and transmission expansion plan required to meet electricity demand, even under drought conditions.

The second part involves scenario analysis using the Global Electrification Platform, which is developed from the OnSSET model, to assess the technologies needed for achieving universal electrification. This includes estimating the population expected to gain access, the additional capacity required for the grid, and the overall investment needed. Various scenarios will be explored to identify the most effective electrification strategy.

A similar approach was applied in Kenya, where OnSSET and OSeMOSYS were used for electrification planning to capture spatial and temporal dynamics in electricity planning [23]. Fig. 3 summarizes the methodology.

3. PyPSA-Earth model development and validation

The development and validation of the PyPSA-Earth model for Zambia are crucial steps in ensuring the accuracy and reliability of energy system simulations. While PyPSA-Earth provides a strong foundation with its global datasets, these must be refined with region-specific data to improve precision. This section outlines the process of developing Zambia's PyPSA-Earth model and validating its key components to ensure its suitability for analyzing Zambia's power system.

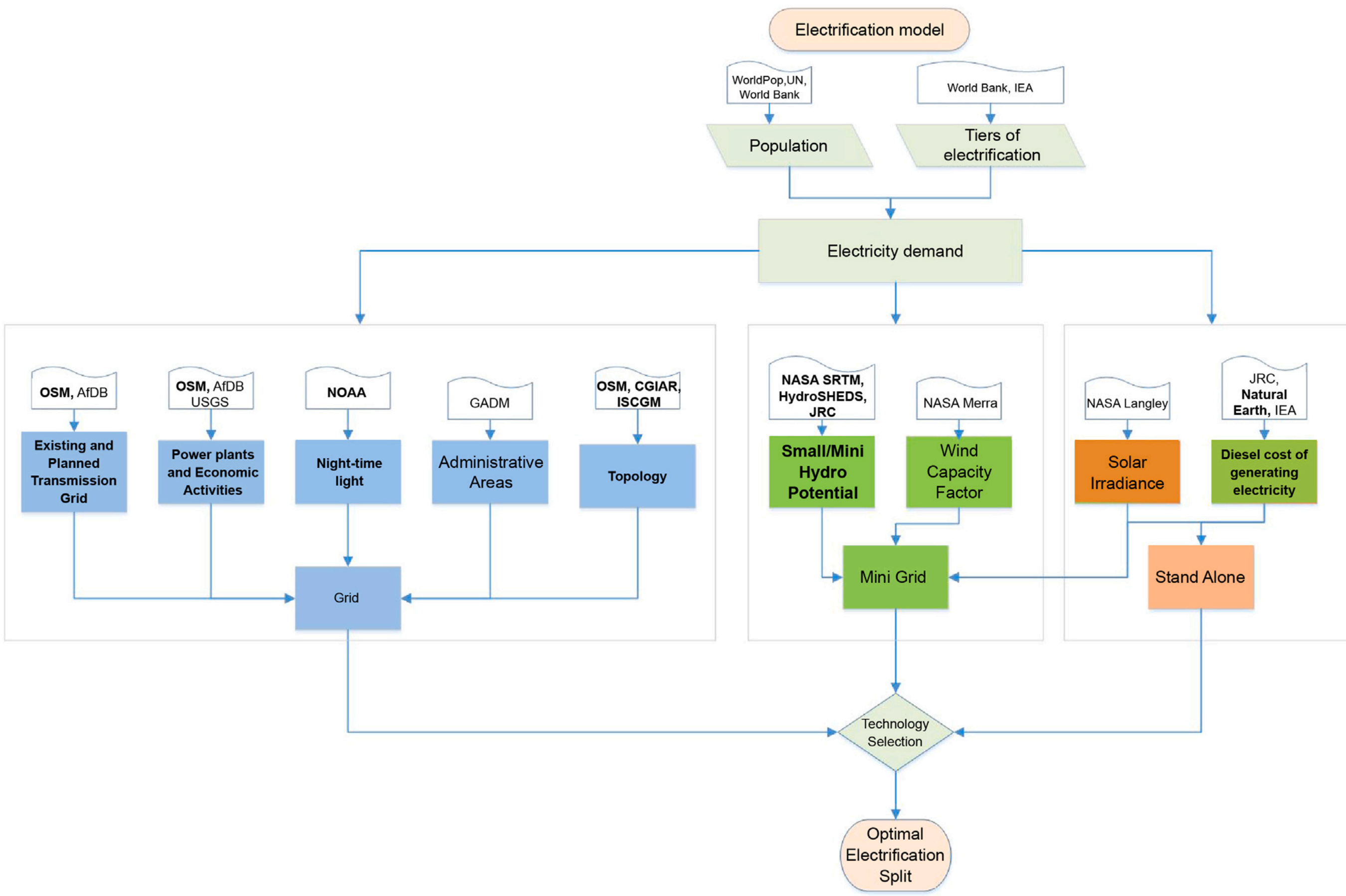


Fig. 2. OnSSet methodology. Source: [13].

3.1. Developing the PyPSA-Earth model for Zambia

PyPSA-Earth's global data repositories simplify the development of an initial working model for any region, including Zambia. This facilitates quick setups even with limited knowledge of the PyPSA-Earth ecosystem and allows for the reuse of work developed for other regions. However, global datasets often lack accuracy or completeness in specific areas, necessitating validation and the integration of external or manually curated data for reliable analysis.

For Zambia, the PyPSA-Earth model required creating and working with the pypsa-zm-data repository, a dedicated repository for Zambia's energy data [55]. Validation tools provided in the PyPSA-Earth ecosystem, such as Jupyter notebooks, assist in identifying and addressing these gaps [44].

3.2. Validation of the PyPSA-Earth model for Zambia

In this section, a summary of the validation of the PyPSA-Earth Zambia is provided. A more detailed description of this model validation is found in a separate report by the authors [56]. The PyPSA-Earth model validation for Zambia was conducted in the following areas of the model:

- Network Topology and Length.
- Power Plant Database.
- Electricity Demand.
- Renewable Potential (Solar and Wind).
- Solar and Wind Capacity Factor.
- Hydropower Inflow Validation.

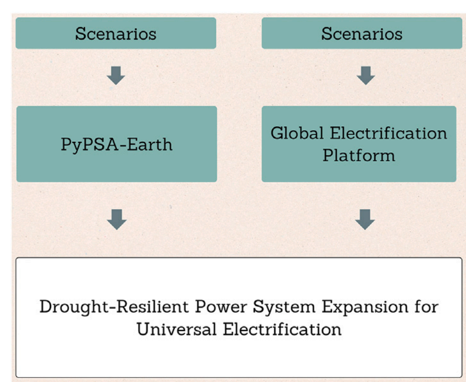


Fig. 3. Proposed research methodology. Source: own creation using [12,46].

The Network Topology and Length validation consisted of visually inspecting the shape of the network and placement of nodes, and a comparison of the network length at every voltage level. The external sources used to validate it were the Zambian Power System Network from ZESCO and the Energy Access Explorer [57], with great similarity to the one in the Energy Access Explorer. Moreover, the total line length discrepancy was found to be less than 7 % with the Zambian Power System Network. As such, the network in the PyPSA-Earth model for Zambia is accurate enough for the study's purposes and hence no changes are required. Fig. A.1 shows the network topology extracted from PyPSA-Earth.

The Power Plant Database validation consisted of comparing the power plants in the PyPSA-Earth model for Zambia with the information in the latest Energy Sector Report by the Energy Regulation Board [8]. Significant discrepancies in the available hydropower capacity were found along with other discrepancies in a few small power plants. The first was caused by the fact that the Kariba Dam is shared between Zambia and Zimbabwe, where each country effectively operates a fixed share of the total installed capacity. PyPSA-Earth reports its total installed capacity as 2,130 MW, but in a Zambian model, only the capacity that is available to Zambia (1080 MW) should be considered. The other discrepancies were possibly caused by referring to the same power plant with different names or simply inaccuracies. In light of this, it would be appropriate to use the Energy Regulation Board database [8].

The Electricity Demand validation consisted of comparing the PyPSA-Earth model for Zambia's estimated electricity demand in 2030 with external sources. PyPSA-Earth uses demand from Global-Energy GIS (GEGIS) [58]. GEGIS produces hourly demand time series by applying machine learning methods using predictors such as temperature, population, GDP, industrial structure, heating, cooling technologies, among others [58]. Based on present demand, expected population growth and GDP growth, the model estimates Zambian electricity demand in 2030 to be 19.19 TWh. This represents a small increase from a total electricity demand of 16.69 TWh in 2020, as reported in the Integrated Resource Plan [59], and is aligned with demand increases in recent years. The Integrated Resource Plan estimates electricity demand in 2030 to be 41.93 TWh, driven by significant transformations in the agricultural, mining, and residential sectors, as well as exports. In light of the huge uncertainty on future electricity demand, several scenarios are considered to cover the range of uncertainty.

Demand distribution and granularity are of paramount importance, especially in the context of electricity access in Zambia. Demand is spatially allocated using a Voronoi partitioning method based on the geographic locations of substations, with each Voronoi cell representing the area closest to a substation [60]. Within each cell, demand is distributed using a 60:40 weighting between GDP (as a proxy for industrial demand) and population (for residential demand). This method results in a high-resolution spatial mapping of electricity demand that reflects both socio-economic activity and demographic distribution, enabling detailed network analysis and informed evaluation of electrification strategies. Fig. A.1 illustrates the regional distribution of electricity demand across the transmission network, as derived from the PyPSA-Earth framework.

The Renewable Potential for solar and wind consisted of comparing the renewable potential for these sources in the PyPSA-Earth model with external sources in the literature. The solar potential estimates in the literature are more than 10,000 TWh/year [61], while the PyPSA-Earth solar potential is 950 TWh/year. While this is a significant discrepancy, it places PyPSA-Earth on the conservative side of the spectrum. Hence, it is robust to use the conservative PyPSA-Earth solar potential estimate. Regarding wind potential for Zambia, PyPSA-Earth estimates it at 1,000 TWh/year. The Renewable Energy Wind Mapping for Zambia estimates annual wind generation if wind farms were installed at several locations, but it does not provide a total estimate for the country. In the absence of such potential estimates in the literature, the PyPSA-Earth default estimate is retained. In summary, no modifications were made to the PyPSA-Earth default renewable potential.

The Solar and Wind Capacity Factor validation consisted of comparing capacity factor figures from the literature with those present in the PyPSA-Earth model for Zambia. The comparison was made with data available from Renewables Ninja [62]. PyPSA-Earth's default capacity factors were slightly lower than those provided by Renewables Ninja, but the differences were below 1 %. Hence, no modifications were needed to the solar and wind time series.

The Hydropower Inflow Validation consisted of comparing water inflows at the nodes of the PyPSA-Earth network with historical data. The PyPSA-Earth model in Zambia estimates water inflows sufficient to generate around 9 TWh of electricity annually, while the Energy Sector Reports show hydropower generation of around 13 TWh in 2019 and 2020, 16 TWh in 2021 and 17 TWh in 2022 and 2023 [8,63–66]. This indicates that the PyPSA-Earth estimate is quite pessimistic, as actual hydropower generation has been significantly above 9 TWh in the last 5 years. Accordingly, this figure needs to be changed and variations in water availability will be incorporated with the introduction of scenarios. In a conservative approach, the default water availability is set to 13 TWh.

4. Energy scenarios

The analysis will focus on two key scenario categories: capacity and transmission expansion scenarios and electrification scenarios. The capacity and transmission expansion scenarios will be modeled using PyPSA-Earth, while the electrification scenarios will be analyzed using the Global Electrification Platform.

4.1. Generation and transmission capacity expansion scenarios

To explore pathways for Zambia's energy transition, the study develops twelve scenarios using the PyPSA-Earth Zambia framework. These scenarios integrate three dimensions: electricity demand growth, water availability, and coal expansion. Each scenario has been carefully designed to address specific research questions, offering a structured framework for evaluating Zambia's transition toward a resilient and low-carbon energy future.

In order to achieve universal electricity access from current levels of 51 % electricity access, the Integrated Resource Plan of Zambia estimates electricity demand in 2030 to be 41.93 TWh, representing approximately 3.4 million new customer connections to the grid [59,67].[1] As such, the study considers three electricity demand scenarios.

The Business-as-Usual (BAU) represents a scenario with little progress toward universal electricity access, keeping the access rate constant at its current level of 51 %. Demand grows in BAU only due to moderate increases in GDP and population. The Significant Electricity Access Growth (SEAG) scenario assumes electricity access rises from 51 % in 2023 to 75 % by 2030, with additional demand from both access expansion and economic and population growth. The High Electricity Access Growth (HEAG) scenario represents an ambitious push for universal electricity access by 2030, where electricity demand grows at twice the historical rate to reflect the impact of full electrification. These scenarios are modeled in PyPSA-Earth Zambia by scaling past electricity use according to projected demand and access levels.

Water availability is a critical factor in this study, addressed through two hydrological scenarios to capture the variability in hydropower generation. In recent years, hydropower output in Zambia has ranged between 13 and 17 TWh annually [66]. To ensure a realistic assessment, long-term river inflow patterns based on historical data are used to model hydropower potential. The Normal Year (NY) scenario reflects average hydrological conditions, corresponding to a total annual generation of 13 TWh. In contrast, the Below Normal Year (BNY) scenario simulates reduced water availability, assuming 67 % of the average inflows. This assumption is in line with previous research by the authors, where below-average years were found to result in similarly low water levels [56].[2] This is also supported by findings in the literature that report such massive declines in annual inflows due to climate variability and other factors affecting hydropower availability [68–70].

The coal expansion dimension includes two scenarios. The Coal Expansion (C) scenario models coal as an extendable carrier, allowing new coal capacity to be added within its local resource potential. This

[1] Customer connection means connections to the medium and low voltage levels 11 kV and 400 V.

[2] The reader is referred to Figs. 6–8 in [56].

Table 1
Generation and transmission capacity expansion scenario summary.

Scenario code	Electricity demand	Water availability	Coal expansion	Description
BAU-NY-NC	Business-as-Usual	Normal Year	No	Modest growth, resilient on avg. hydrological cond., no coal expansion.
BAU-BNY-NC	Business-as-Usual	Below Normal Year (x0.67)	No	Modest growth, resilient on below avg. hydrological cond, no coal expansion.
SEAG-NY-NC	Signif. Electricity Access Growth (x1.5)	Normal Year	No	Signif. growth, resilient on avg. hydrological cond, no coal expansion.
SEAG-BNY-NC	Signif. Electricity Access Growth (x1.5)	Below Normal Year (x0.67)	No	Signif. growth, resilient on below average. hydrological cond., no coal expansion.
HEAG-NY-NC	High Electricity Access Growth (x2)	Normal Year	No	Universal access, resilient on avg. hydrological cond., no coal expansion.
HEAG-BNY-NC	High Electricity Access Growth (x2)	Below Normal Year (x0.67)	No	Universal access, resilient on below avg. hydrological cond., no coal expansion **(critical)**.
BAU-NY-C	Business-as-Usual	Normal Year	Yes	Modest growth, resilient on average. hydrological cond., coal expansion.
BAU-BNY-C	Business-as-Usual	Below Normal Year (x0.67)	Yes	Modest growth, resilient on below average. hydrological cond., coal expansion.
SEAG-NY-C	Signif. Electricity Access Growth (x1.5)	Normal Year	Yes	Signif. growth, resilient on average. hydrological cond., coal expansion.
SEAG-BNY-C	Signif. Electricity Access Growth (x1.5)	Below Normal Year (x0.67)	Yes	Signif. growth, resilient on below avg. hydrological cond., coal expansion.
HEAG-NY-C	High Electricity Access Growth (x2)	Normal Year	Yes	Universal access, resilient on avg. hydrological cond., coal expansion.
HEAG-BNY-C	High Electricity Access Growth (x2)	Below Normal Year (x0.67)	Yes	Universal access, resilient on below avg. hydrological cond., coal expansion **(critical)**.

requires manually adding coal power plants with zero installed capacity in the base year across all nodes, as PyPSA-Earth only allows expansion where coal power plants already exist. The No Coal Expansion (NC) scenario maintains coal as a non-extendable carrier, as per the PyPSA-Earth default configuration.

The combination of these dimensions results in twelve integrated scenarios, as shown in Table 1.

4.2. Electrification scenarios

The electrification scenarios will include the following cases:

- **Business as Usual (BAU) – All Technologies Activated (SN1):** This scenario considers grid extension, mini-grids, and standalone systems as potential electrification options. It reflects the current approach, integrating multiple technologies to meet diverse needs across urban and rural areas.
- **50 % PV Cost Reduction (SN2) – With and Without Hydro Mini-Grids:** This explores the impact of declining solar PV costs on electrification, with one case allowing hydro mini-grids and another excluding them. This analysis is pertinent given the global trend of declining solar PV costs, which could make solar solutions more accessible and cost-effective for rural electrification.
- **BAU – Grid Connection Only (SN3):** This scenario assumes that electrification efforts rely solely on grid expansion, excluding off-grid solutions. It provides insights into the challenges and costs associated with extending the national grid to remote and rural areas, where infrastructure development can be more complex and expensive.
- **BAU without Solar Home Systems (SHS) (SN4):** This scenario examines the role of SHS in rural electrification, comparing outcomes when they are included versus excluded as an option. The analysis helps determine the impact of excluding SHS on achieving electrification targets and the potential need for alternative solutions.
- **BAU with Residential Demand Only (SN5):** This scenario focuses solely on meeting residential electricity demand. It evaluates the feasibility of achieving electrification targets by prioritizing household connections, which is crucial for improving living standards and supporting socio-economic development.

By integrating the findings from these scenarios, the most cost-effective and sustainable pathways for expanding Zambia's electricity access can be identified while ensuring grid stability and resilience.

4.3. Scenario implementation

The PyPSA-Earth Zambia framework integrates these scenarios through several key steps. First, demand profiles are projected for each scenario, reflecting historical trends and future growth driven by electrification and GDP expansion. Next, hydropower constraints are modeled to simulate availability under typical and drought conditions, using historical inflow data and adjustments for reduced water supply. Renewable integration is then optimized, focusing on deploying solar and wind energy to complement hydropower and sustainably meet demand. This robust implementation ensures that all twelve scenarios are thoroughly analyzed to generate actionable insights for Zambia's energy transition.

For the electrification scenarios, the Global Electrification Platform was used. The platform's scenario section and filter section were configured accordingly, with the appropriate levers adjusted to meet the specific requirements of each scenario [54].

4.4. Research questions and energy scenarios alignment

Each scenario has been carefully designed to address specific research questions, offering a structured framework for evaluating Zambia's transition toward a resilient and low-carbon energy future. Below is a summary of how the scenarios align with the first three research questions.

4.4.1. Research question 1: reducing hydropower dependency while ensuring energy security and sustainability by 2030

This research question investigates strategies for decreasing Zambia's reliance on hydropower while maintaining a secure and sustainable energy supply. The relevant scenarios include BAU-BNY-NC, BAU-BNY-C, HEAG-BNY-NC, and HEAG-BNY-C. These scenarios assess the resilience of Zambia's power system under conditions of reduced water availability, highlighting vulnerabilities associated with over-reliance on hydropower without diversification.

In particular, the HEAG-BNY scenarios evaluate the role of alternative renewable energy sources such as solar and wind in maintaining grid stability during severe droughts.

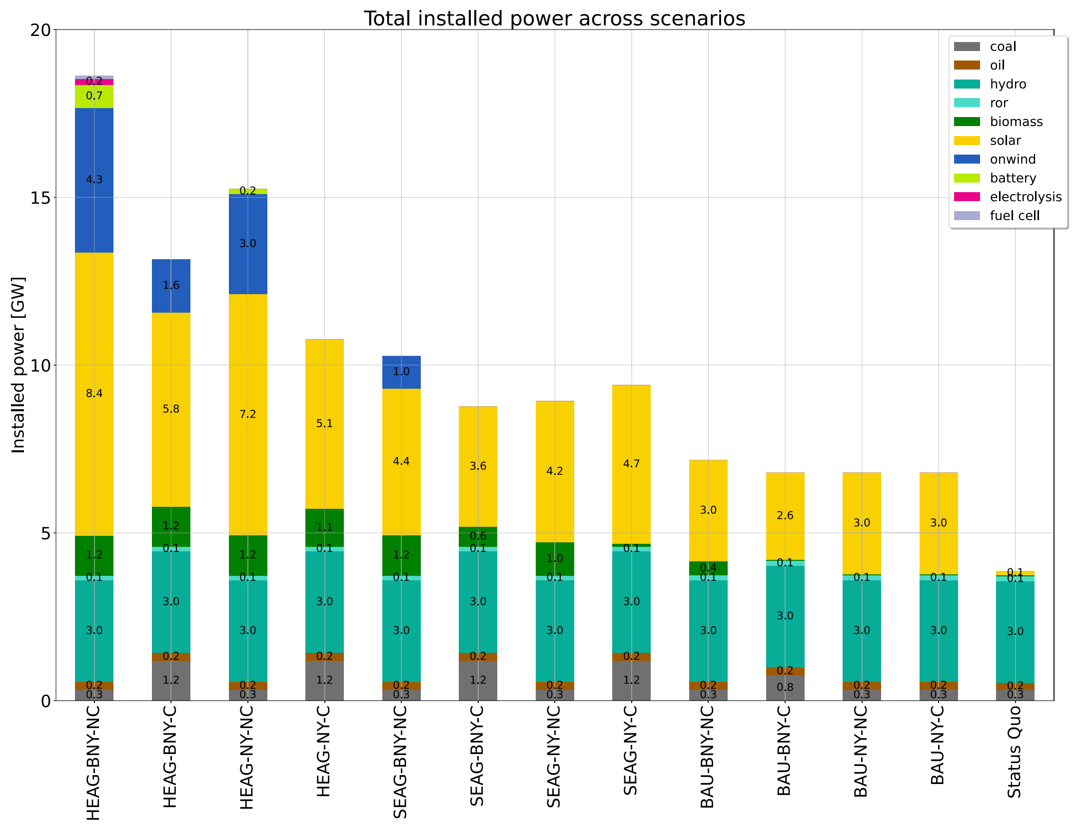


Fig. 4. Installed capacity in 2030 in GW across the considered scenarios. The figure shows the cumulative installed capacities across scenarios, which contains current power plants plus new investments from the scenarios. Data on the status quo are from [66].

4.4.2. Research question 2: trade-offs between universal electricity access, renewable energy integration, and energy security under varying hydrological conditions

This research question examines the balance between expanding electricity access, integrating renewable energy, and maintaining grid security under different hydrological conditions. The relevant scenarios include SEAG-BNY-NC, SEAG-BNY-C, HEAG-NY-NC, and HEAG-NY-C. These scenarios explore how increased electricity demand interacts with reduced water availability, exposing potential supply vulnerabilities.

Furthermore, they highlight how universal electricity access can be achieved under favorable hydrological conditions by scaling up renewable energy integration.

4.4.3. Research question 3: policies, technologies, and investment frameworks for Zambia's low-carbon, resilient energy transition

This research question focuses on identifying effective policies, technological advancements, and investment strategies that can facilitate Zambia's transition to a low-carbon and resilient energy system. The relevant scenarios include HEAG-NY-NC, HEAG-NY-C, HEAG-BNY-NC, and HEAG-BNY-C.

These scenarios assess the effectiveness of policy support and technological interventions in achieving universal electricity access and integrating renewables under both normal and extreme hydrological conditions.

5. Results and analysis

This section presents the key findings from the analysis of Zambia's power system, focusing on two critical aspects: generation and transmission capacity, and electrification. The generation and transmission analysis evaluates the system's ability to meet demand under varying hydrological conditions, highlighting the role of different energy sources, storage solutions, and grid flexibility. The electrification analysis examines strategies for expanding electricity access, considering grid extension, mini-grids, and standalone systems to achieve universal access in a cost-effective manner. The results provide insights into system resilience, investment priorities, and policy recommendations for a sustainable energy future.

5.1. Generation and capacity expansion planning

We aim to provide insights into the energy landscape in Zambia as well as the impact of three dimensions discussed in the scenarios settings section on the construction of the energy system in Zambia. First, the main findings in all scenarios are summarized, elaborating on certain interesting hours to show the seasonality of the energy vectors in Zambia and how the system maintains reliability and security of supply. Fig. 4 shows a considerable investment in solar energy.

Across all scenarios, solar energy dominates as a key investment due to its vast potential and cost-effectiveness. Solar energy plays a pivotal role in enhancing the country's electricity sector resilience and is consistently integrated into the system, even under low demand growth scenarios. For example, in low electricity access scenarios with normal water availability, at least 3 GW of solar power is installed.

Higher electricity access scenarios (HEAG) drive the largest investments in renewables, particularly solar and wind technologies. The demand for increased electricity access acts as the main trigger for these investments and capacity expansions. Biomass potential is fully utilized in these scenarios, while the absence of coal investments or lower water availability further stimulates wind and solar development. Moreover, scenarios with high electricity access (HEAG) and no coal extension (nc) are the only scenarios where battery storage investments are triggered by the system to balance supply in the absence of dispatchable

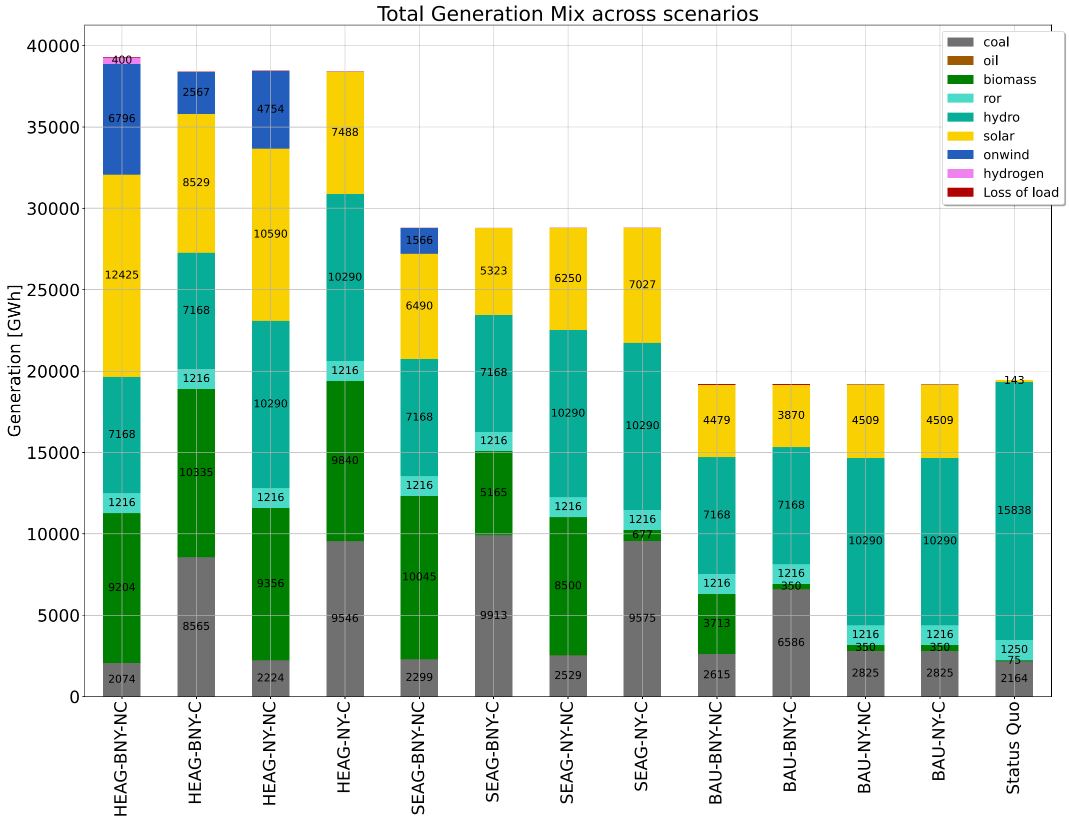


Fig. 5. Power generation mix in 2030 in GWh. The figure shows the cumulative generation across scenarios. Data on the status quo are from [66].

coal generation. Low water availability (BNY) also leads to more battery investments, as hydropower is flexible but limited.

Coal investments emerge as a preferred choice in scenarios where coal is permitted (c) due to its dispatchability and lower operational and investment costs compared to biomass. For example, in the HEAG-BNY-C scenario, 900 MW of coal power was added to secure supply, unlike the HEAG-NY-NC scenario where the absence of coal led to increased investments in both wind and solar technologies. This trend holds true in all scenarios where coal investments are allowed.

Water availability has a significant impact on Zambia's energy system. In moderate electricity access scenarios (SEAG), low water availability (BNY) without coal investments (nc) leads to higher investments in wind energy (1000 MW) and full utilization of biomass potential (1193 MW). On the other hand, with coal investments (c) under the same conditions, solar (800 MW) and biomass (600 MW) investments are lower, with no wind investments, while coal capacity increases by 900 MW.

Fig. 5 shows the energy mix of all scenarios. High electricity access scenarios highlight a significant role for renewables in the energy mix. For example, at least 20 % of the energy mix will come from intermittent renewables like solar and wind under the HEAG-NY-C scenario. This share increases to 50 % in the HEAG-BNY-NC scenario and further rises to 92 % when hydro and biomass energy are included. However, coal expansion significantly impacts the energy mix. In the HEAG-BNY-C scenario, coal generation triples compared to HEAG-BNY-NC, reducing the share of intermittent renewables to 29 %. Despite this, the HEAG-BNY-NC scenario faces a 15 GWh loss of load over the year, raising concerns about balancing energy security with sustainability.

In moderate electricity access scenarios, hydropower availability strongly influences dispatchable energy sources, mainly coal and biomass. Low water availability increases reliance on coal and biomass, with coal preferred due to its lower operational cost. Conversely, normal water availability reduces coal dependence, allowing for a more balanced energy mix.

In low electricity access scenarios, coal expansion occurs only under conditions of low water availability. Here, a higher coal share directly suppresses solar's contribution to the energy mix, highlighting coal's role in compensating for reduced hydro resources at the expense of renewable integration.

Hydropower remains a cornerstone of Zambia's energy mix. Under normal water availability, hydro contributes 10,290 GWh annually, whereas low water availability reduces this to 7,168 GWh, emphasizing hydro's importance despite its climate vulnerability. In scenarios with low hydro availability, solar energy compensates for the hydro deficit. Zambia's energy future will rely on hydropower, complemented by solar and wind technologies, and supported by dispatchable sources like biomass and coal. This combination ensures the country meets its energy needs while enhancing power system resilience against climate hazards.

5.2. System costs and CO2 emissions

The previous results can be directly translated into system costs and cumulative emissions. Fig. 6 illustrates the total emissions across all scenarios. Scenarios without coal expansion have the lowest emissions, as coal remains the primary emissions driver in Zambia's energy system. In contrast, scenarios permitting coal expansion, especially under higher electricity access, show significantly increased emissions.

For instance, under low water availability, low electricity access results in cumulative emissions of 5,025 thousand tCO_2, while moderate electricity access causes a sharp rise to 7,563 thousand tCO_2. Interestingly, high electricity access leads to emissions of 6,534 thousand tCO_2. Despite increased energy demand, this remains lower than the moderate access scenario, highlighting the role of renewables in offsetting emissions growth.

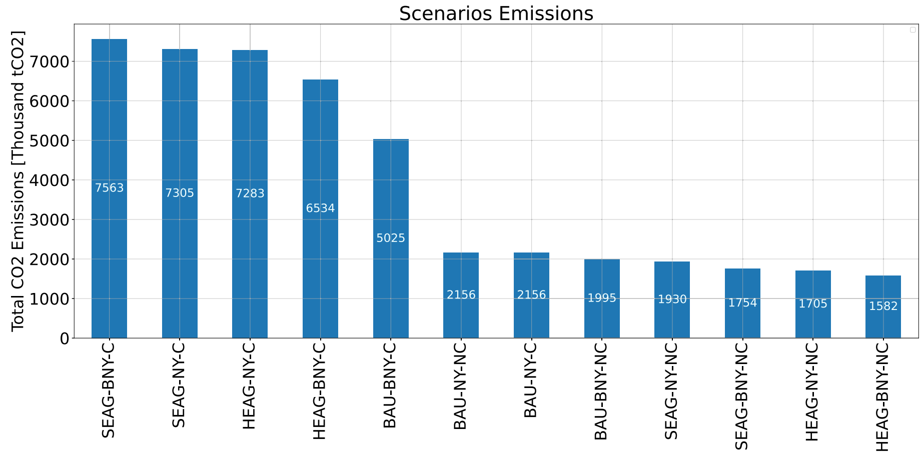


Fig. 6. Total emissions per scenario in 2030 in thousand tonnes of CO2.

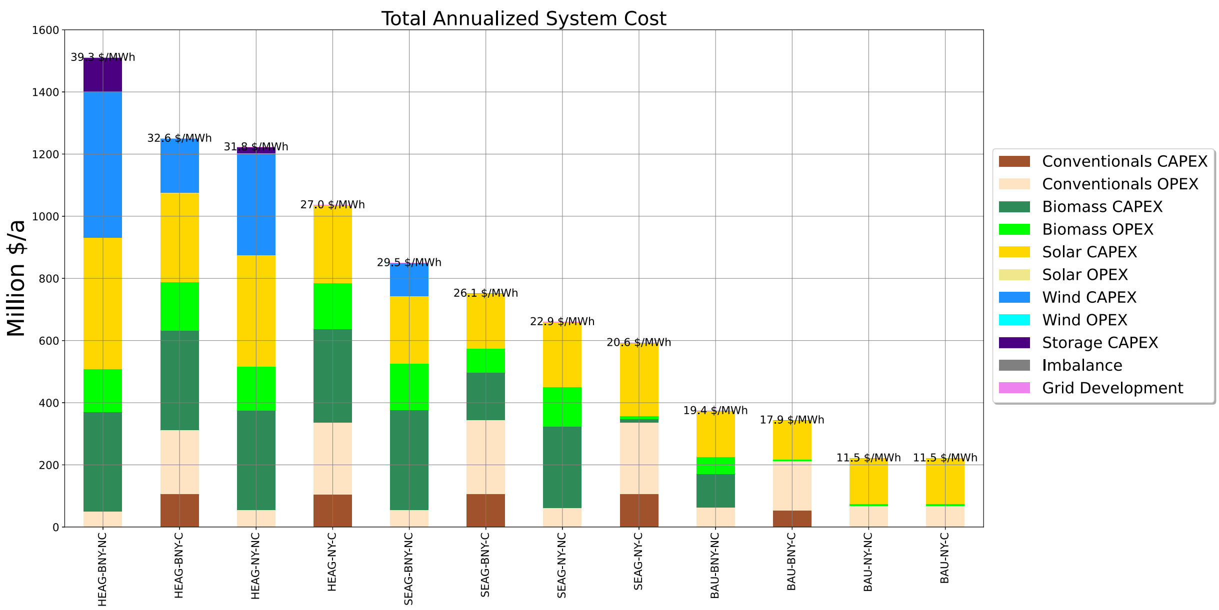


Fig. 7. Total annualised system cost per scenario [Million Dollars/Year].

This trend highlights an intriguing balance: moderate electricity access scenarios show a mix of renewable and non-renewable generation, as previously demonstrated in Fig. 5. In contrast, high electricity access scenarios drive further investments in renewables, which offset some of the emissions caused by increased energy demand. This result underlines the critical role of renewable energy in mitigating emissions while supporting Zambia's growing energy needs.

In terms of costs, Fig. 7 highlights the key trends driving system cost contributions. Firstly, scenarios with high electricity access consistently result in higher cumulative annual system costs. Secondly, lower water availability increases costs due to reduced reliance on hydropower, which necessitates alternative investments. Lastly, stopping coal expansion leads to higher system costs, as the system requires additional capital investments in renewables and other energy sources to meet demand.

The transition toward intermittent renewable energy resources involves relatively higher upfront costs compared to scenarios relying on coal. However, in the long term, systems with a higher share of fossil-fuel power plants prove to be more expensive due to higher operational costs, whereas renewables offer significant economic advantages. For instance, in low electricity access scenarios with low water availability, the BAU-BNY-NC scenario, which excludes coal expansion, requires 254 million USD/year for capital investments in solar and biomass, while the BAU-BNY-C scenario requires only 178 million USD/year for coal and solar investments. However, the operational costs in the renewables-dominated system (BAU-BNY-NC) are approximately 119 million USD/year, nearly 45 million USD/year less than the coal-based scenario. This demonstrates that while capital expenditures for renewables are higher initially, the lower operational costs make renewables more economically advantageous in the long run.

This difference becomes even more pronounced in higher electricity access scenarios. For example, the HEAG-BNY-NC scenario requires 1,322 million USD/year in capital expenditures for investments in renewables such as solar, wind, and biomass, compared to 888 million USD/year in the HEAG-BNY-C scenario, which allows coal expansion. However, the operational costs in the HEAG-BNY-NC scenario are significantly lower at 188 million USD/year, nearly half of the operational costs in the coal-dominated system. Over time, investments in renewable energy technologies drive overall system costs down while simultaneously reducing emissions, leading to a cost-optimal energy system

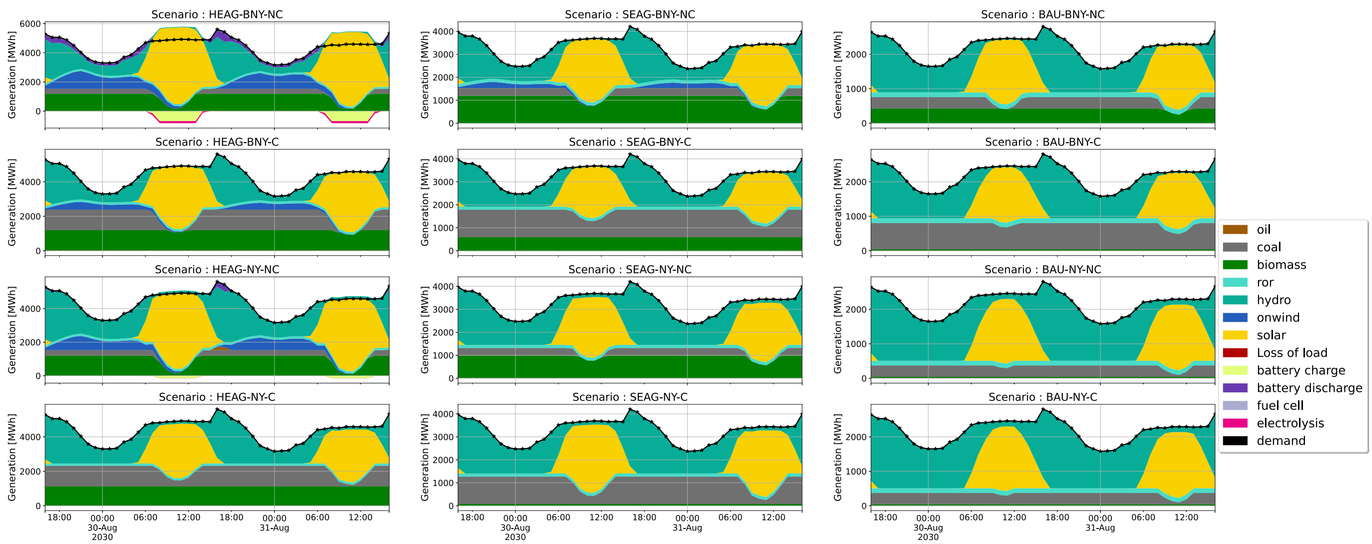


Fig. 8. Dispatch with seasonal peak demand in 2030.

powered predominantly by renewables. This underscores the long-term economic and environmental benefits of prioritizing renewables expansion over-reliance on fossil fuels.

5.3. Resilience of future energy systems

The dispatch analysis across scenarios highlights the dynamic interplay of various generation assets, particularly during periods of peak demand (Fig. 8), maximum hydro generation (Fig. 9), and maximum intermittent renewable infeed (Fig. 10). For instance, Fig. 8 shows the electricity dispatch at the peak demand time, providing insights into how different technologies contribute to meeting the system's peak demand requirements.

In all scenarios, peak demand occurs during early evening hours, as shown by sharp rises in the demand curve. Solar generation dominates during daylight but drops to zero after sunset, creating a demand gap that must be filled by other technologies to maintain system stability. In scenarios with low water availability and no coal expansion (BNY-NC) shown in the 1st column, biomass plays a crucial role in meeting demand, while in high electricity access scenarios (HEAG), batteries become essential by storing solar energy during peak hours and discharging during peak demand.

In coal expansion scenarios (2nd column), coal provides steady dispatchable generation but ramps down when solar is available. This trend is consistent across all scenarios. Coal expansion reduces the need for solar and battery investments due to its dispatchability, whereas scenarios without coal rely more on biomass, wind, and batteries to balance the system under higher electricity access.

Scenarios with normal water availability (3rd and 4th columns) experience fewer extreme events, as hydro generation plays a more significant role during peak demand, reducing reliance on biomass and backup resources. This is further illustrated in Fig. 9, which shows the hour with maximum hydro generation.

During these periods, hydro plays a crucial role in meeting demand, especially when solar is unavailable at night or in the early morning. As a reliable, dispatchable resource, hydro helps stabilize the system by filling the gaps left by intermittent renewables like solar and wind, reducing reliance on fossil fuels and biomass.

In scenarios without coal expansion (1st and 3rd columns), hydro becomes even more critical, compensating for the lack of steady coal power. Its consistent output during peak hours minimizes the need for costly oil-based generation and reduces dependence on coal and biomass. However, this reliance highlights the system's vulnerability to water availability, particularly in low-water scenarios.

Balancing hydro with other renewables like solar and biomass is essential for long-term resilience and energy security. This dynamic is further illustrated in Fig. 10, which shows the hour with maximum intermittent renewables infeed alongside minimum hydro infeed, particularly in the extreme scenario HEAG-BNY-NC (high electricity access, low water availability, and no coal expansion).

During these periods, solar generation and, to a lesser extent, wind dominate, demonstrating the capacity of intermittent renewables to meet significant portions of the system demand during peak sunlight hours. In contrast, hydro generation is reduced to minimal levels, highlighting its role as a flexible, dispatchable resource that can scale back when generation from renewable sources is abundant. This interaction underscores the importance of system flexibility in integrating high shares of intermittent renewables. While solar and wind effectively contribute during their peak generation times, hydro resources act as a balancing mechanism, stepping in during periods of reduced renewable availability (e.g., nighttime or cloudy conditions). The reduced hydro generation during these peak renewable hours conserves water for later use, ensuring system reliability when renewables decline.

5.4. Analysis of electrification scenarios in Zambia

This section analyzes different electrification scenarios for Zambia, assessing their impact on capacity addition, population connected by 2030, and the investment required. The scenarios examined include SN1, which considers grid extension, mini-grids, and standalone systems. SN2 explores the impact of declining solar PV costs with and without hydro mini-grids. SN3 assumes electrification efforts rely solely on grid expansion, excluding off-grid solutions. SN4 examines the role of solar home systems in rural electrification by comparing outcomes when they are included versus excluded as an option. Lastly, SN5 focuses solely on meeting residential electricity demand, excluding industrial and commercial consumption.

5.4.1. Capacity added

The capacity added across scenarios varies significantly depending on the electrification strategy employed. SN1 and SN2 exhibit the highest capacity additions, reaching 1,095.1 MW and 1,064 MW, respectively. These scenarios rely on large-scale investments in both grid connections and decentralized solutions, such as solar home systems and mini-grids. In contrast, SN3, SN4, and SN5 show lower capacity

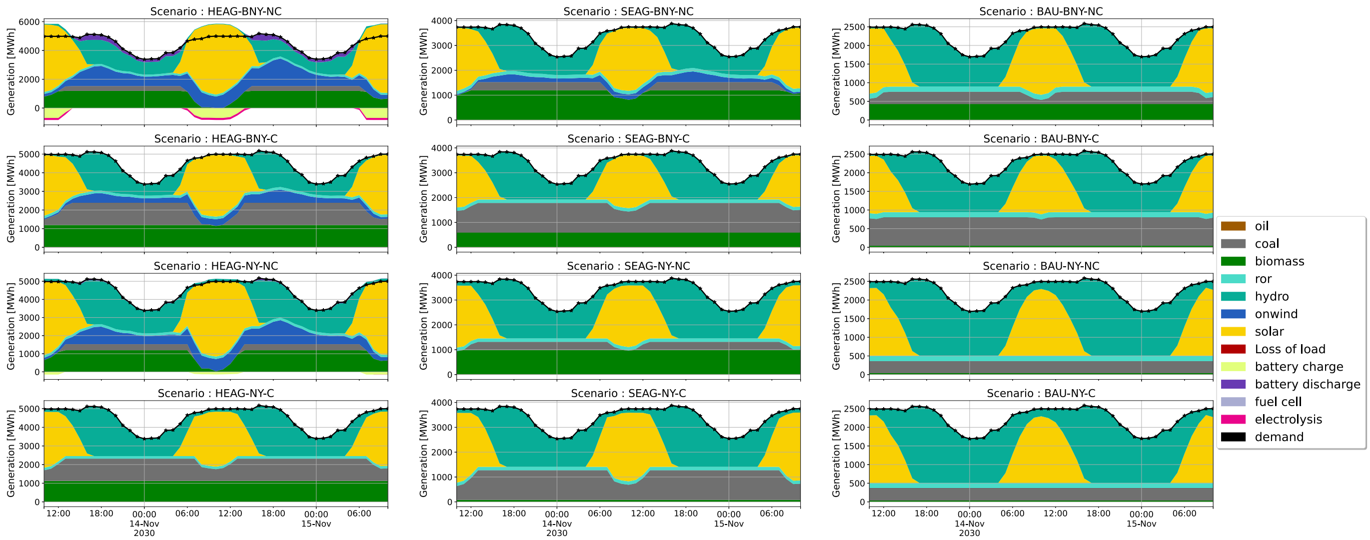


Fig. 9. Dispatch with maximum hydro in 2030.

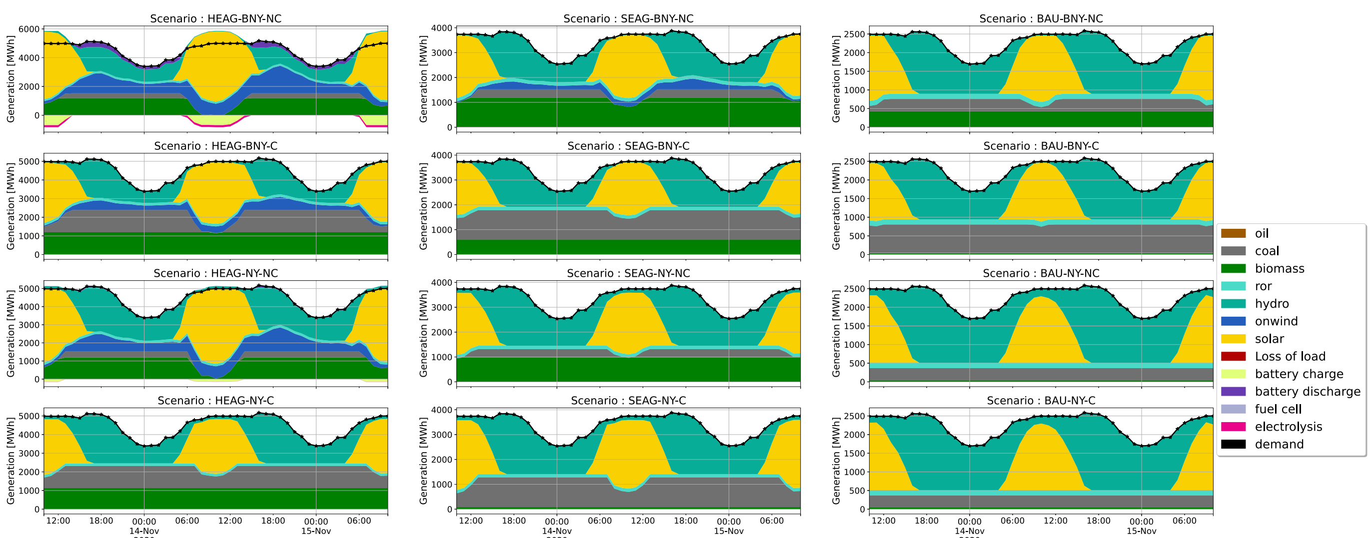


Fig. 10. Dispatch with maximum intermittent renewable infeed.

additions, with 720 MW, 784 MW, and 713 MW, respectively, as they focus more on grid-based electrification with limited reliance on off-grid solutions. Solar home systems contribute significantly to capacity expansion, particularly in SN1 and SN2, where they form a substantial portion of the total. However, grid connections remain the dominant expansion method across all scenarios, particularly in those prioritizing nationwide electrification. See Fig. 11(a).

5.4.2. Population connected to electricity by 2030

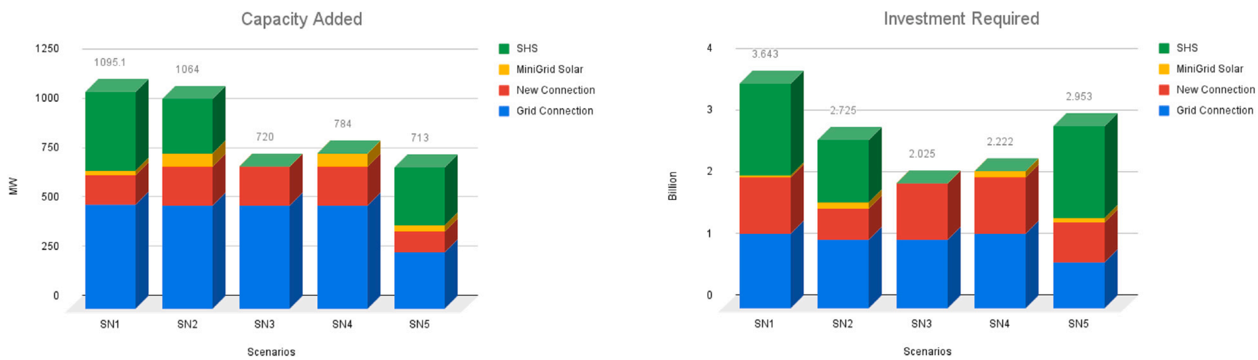
The number of people gaining access to electricity varies across scenarios. SN1 and SN2 achieve the highest electrification rates, connecting 24.278 million and 24.629 million people, respectively. These scenarios employ a balanced mix of grid extensions, new connections, and off-grid solutions. SN3 and SN4 see significantly lower connections, reaching 13.7 million and 14.03 million people, respectively, suggesting a constrained electrification approach due to limited grid expansion or investment. SN5 uniquely reaches 24.54 million people, similar to SN1 and SN2, indicating that a hybrid approach using solar home systems and mini-grid solar can achieve high electrification levels. A key insight from this analysis is that a higher reliance on decentralized solutions allows SN5 to reach similar electrification levels as SN1 and SN2 but with a different investment structure. See Fig. 11(b).

5.4.3. Investment required

Investment requirements for electrification vary considerably across scenarios. SN1 requires the highest investment at 3.643 billion USD due to extensive grid expansion and solar home system deployment. SN2 follows with 2.725 billion USD, reflecting a similar electrification strategy with slightly reduced costs. SN3 and SN4 require the least investment, at 2.025 billion USD and 2.222 billion USD, respectively. These lower-cost scenarios rely on incremental grid expansion and fewer off-grid solutions. SN5 requires 2.953 billion USD, positioning it between high-investment scenarios like SN1 and SN2 and low-investment scenarios like SN3 and SN4, suggesting a cost-effective approach that still achieves high electrification levels. See Fig. 12.

5.5. Key findings

An important dimension of this analysis is the electrification cost per capita, which provides insight into the cost-effectiveness of each scenario. SN2 and SN5 emerge as the most cost-efficient, with per



(a) Capacity Added (b) Investment Required

Fig. 11. Capacity added and investment required across the considered scenarios.

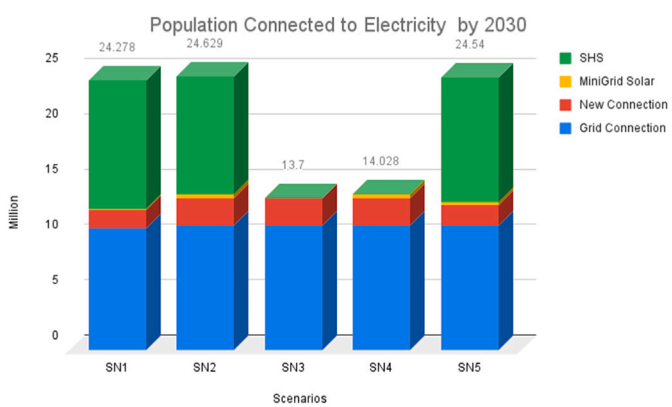


Fig. 12. Population connected to electricity by 2030 across the considered scenarios.

capita costs of approximately $111 and $120, respectively. These scenarios leverage declining solar PV costs and decentralized technologies like solar home systems and mini-grids to connect large populations at relatively lower investments. SN1, despite requiring the highest total investment at $3.643 billion, maintains a competitive per capita cost of around $150 due to its extensive coverage of over 24 million people. In contrast, SN3 and SN4, which connect fewer people, exhibit higher per capita costs of $148 and $158, respectively, reflecting the inefficiencies of relying predominantly on grid-based expansion with limited use of off-grid technologies. These findings underscore the importance of diverse electrification strategies that balance investment with broad and equitable access, highlighting SN2 and SN5 as particularly effective models for achieving universal electrification by 2030.

Achieving universal electrification requires significant capacity additions, as demonstrated by SN1 and SN2, which support full-scale electrification efforts. Decentralized solutions can achieve high electrification at lower costs, as evidenced by SN5, which shows that a combination of solar home systems and mini-grid solar can match electrification levels of SN1 and SN2 with optimized investment. Investment trade-offs are also evident, as high-electrification scenarios such as SN1, SN2, and SN5 require substantial financial input, whereas lower-cost scenarios such as SN3 and SN4 achieve partial electrification with reduced capital requirements.

Achieving universal electricity access by 2030 requires a balanced approach to grid expansion, off-grid solutions, and investment optimization. While large-scale investments, as seen in SN1 and SN2, offer the fastest electrification path, SN5 provides an alternative, cost-effective strategy leveraging solar home systems and mini-grid solar.

6. Policy implications and key takeaways

The analysis provides insights into Zambia's different pathways and offers actionable policy recommendations for its energy transition. Here, we comprehensively answer the research question with key takeaways on the different dimensions of the Zambian energy transition and compare the main outcomes and modeling trends with the local and international literature.

6.1. Hydropower dependency: how can Zambia reduce its hydropower dependency while ensuring energy security and sustainability by 2030?

Despite the massive expansion of intermittent renewable energy in the different pathways for Zambia, hydropower still plays an important role in securing the supply. Fig. 13 shows the role of each technology in covering parts of the demand in future energy systems.

Hydropower continues to play a vital role in securing energy supply, particularly in normal water availability scenarios. However, its prominence diminishes as Zambia moves toward higher levels of electricity access. For example, under low electricity access scenarios, hydropower dominates, covering more than 70 % of the demand for over 5000 hours per year and reaching peaks of 80 % coverage for around 1000 hours annually. However, this figure drops drastically in moderate and high electricity access scenarios. In moderate electricity access scenarios, hydro energy covers less than 60 % of demand for less than 1000 hours a year, and in high electricity access scenarios, it covers less than 200 hours at 60 % of demand. That being said, coal and biomass expansion in the system directly correlates with lowering hydro's role in covering the demand. This is true both on an hourly basis, as shown in 13, or as a total over the year as shown in 5. This reduction highlights the need for alternative energy sources to secure supply and support sustainability goals.

One of the most effective strategies for reducing hydropower dependency is the expansion of solar and wind energy. Solar energy, in particular, emerges as a critical resource in high electricity access scenarios, where it covers over 100 % of demand for 500 hours annually, enabling storage via battery charging and hydrogen production. However, solar generation is not without limitations, as its infeed remains zero for over 4,400 hours annually and contributes less than 10 % of demand for 5,000 hours, a fact that holds true in all scenarios.

Biomass energy plays an increasingly important role in providing dispatchable power, particularly in scenarios without coal expansion. Its flexibility allows it to balance the grid during periods of low solar and wind infeed, reducing the burden on hydropower. In scenarios where coal is allowed to expand, hydropower's role diminishes further, as coal provides a stable and dispatchable source of electricity. However, the environmental trade-offs of increased coal use must be weighed carefully for long-term solutions.

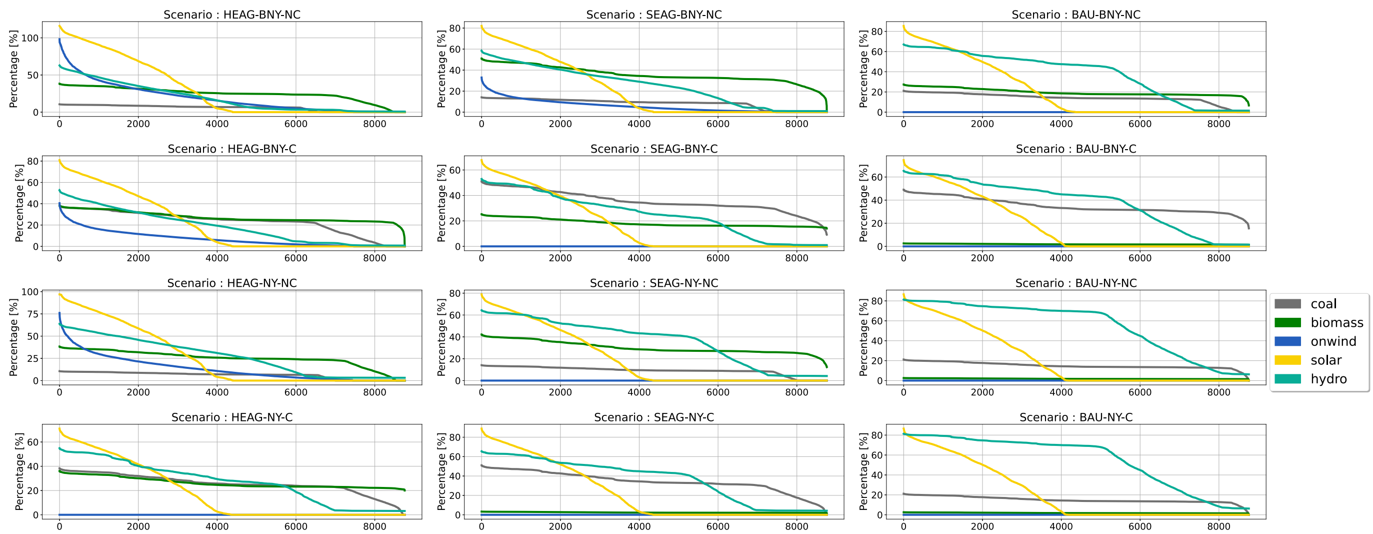


Fig. 13. Generator duration curve in 2030. The values are computed as the dispatch of each technology at each hour as a percentage of the actual demand at that hour.

The same conclusions on hydro's role in future energy systems apply in the case of low water availability. However, the role becomes less important due to the limited hydro capabilities in drought scenarios. For instance, in high electricity access scenarios, hydro covers less than 5 % of the demand on an hourly basis for more than 2800 hours in scenarios with coal expansion (2nd column, 1st raw), and for more than 3500 hours without coal expansion (1st column, 1st raw).

By prioritizing the diversification of its energy mix and investing in flexible technologies, the energy system in Zambia can ensure reliability while simultaneously reducing its reliance on hydropower, as is the case currently. Expanding solar and wind capacity, leveraging biomass as a dispatchable resource, with the supplementation of storage technologies will allow Zambia to meet its growing electricity demand sustainably. The findings are also supported by local literature. Daka and Farzaneh 2023 develop an energy demand–supply framework using an autoregressive statistical model to analyze future low-carbon scenarios in Zambia [71]. Their main finding is that hydro remains dominant in future energy systems. This is also supported by Quesnel 2024, where OSeMOSYS was used to investigate the energy system in Zambia, and found that increasing investments in renewable energy can reduce reliance on hydropower by 30 % [72]. Nyoni 2022 used OSeMOSYS and FlexTool to study the Zambian energy transition until 2063 and highlighted the importance of diversification in achieving climate targets for the country [73].

6.2. What are the trade-offs between achieving universal electricity access, renewable energy integration, and energy security under varying hydrological conditions?

The impact of achieving higher electricity access on the share of hydropower in the energy mix is clearly shown in Fig. 14. In low electricity access scenarios, hydropower dominates the energy mix, contributing over 50 % in normal water availability and more than 37 % in low water availability. This heavy reliance on hydro highlights its role as a cornerstone of energy security, given its reliability and flexibility. However, this reliance also reveals a vulnerability: under low water availability conditions, the system is constrained, and the reliance on hydro diminishes, making the system less resilient. While hydro ensures renewable energy integration in these scenarios, the limited demand reduces the pressure to diversify the energy mix or incorporate other technologies.

As electricity access expands to moderate levels, the energy system begins to diversify. Hydro's share in the energy mix decreases to approximately 36 % under normal water conditions and to 25 % under low water availability. This shift highlights the trade-off between increasing electricity access and hydro's dominance in the system. With higher access, the system increasingly depends on intermittent renewables, such as solar and wind, whose integration reduces reliance on hydro. However, this diversification introduces challenges to energy security, as the system must now manage the variability of these renewables. The role of flexible resources like biomass and coal grows, mitigating some of these challenges but also raising concerns about long-term sustainability and resource availability.

The trade-offs become even more pronounced in high electricity access scenarios. Hydro's share falls drastically, contributing just 27 % under normal water availability and less than 20 % under low water availability. This sharp decline underscores the system's shift toward massive solar and wind integration, coupled with increased dependence on biomass and coal to provide the necessary flexibility. The reliance on coal, in particular, reflects a trade-off between achieving universal electricity access and maintaining environmental sustainability. While coal offers dispatchable generation and enhances energy security, it comes at the cost of higher emissions. Meanwhile, biomass becomes essential for balancing the system, though its scalability may pose long-term challenges.

Under low water availability scenarios, the trade-offs are most apparent. Hydro's reduced role places significant pressure on other energy resources to maintain energy security. While renewables like solar and wind dominate during their peak periods, their variability underscores the need for dispatchable resources to stabilize the grid. This situation emphasizes the importance of system flexibility and resilience, achieved through a combination of storage technologies, dispatchable renewables (biomass), and even fossil-based resources like coal in some scenarios.

6.3. What policies, technologies, and investment frameworks best support Zambia's transition to a low-carbon, resilient energy system?

To support Zambia's transition to a low-carbon and resilient energy system, policies should prioritize diversifying the energy mix, scaling up renewable energy, and enhancing system flexibility. Overall, the study shows that a diverse generation mix, leveraging the strengths of hydro, solar, wind, biomass, and batteries, is critical for achieving energy security and decarbonization. Hydro's flexibility and dispatchability stabilize the grid, supporting higher renewable penetration and reducing dependence on coal. However, the system's reliance on hydro also calls for strategies to enhance water resilience, particularly under scenarios of

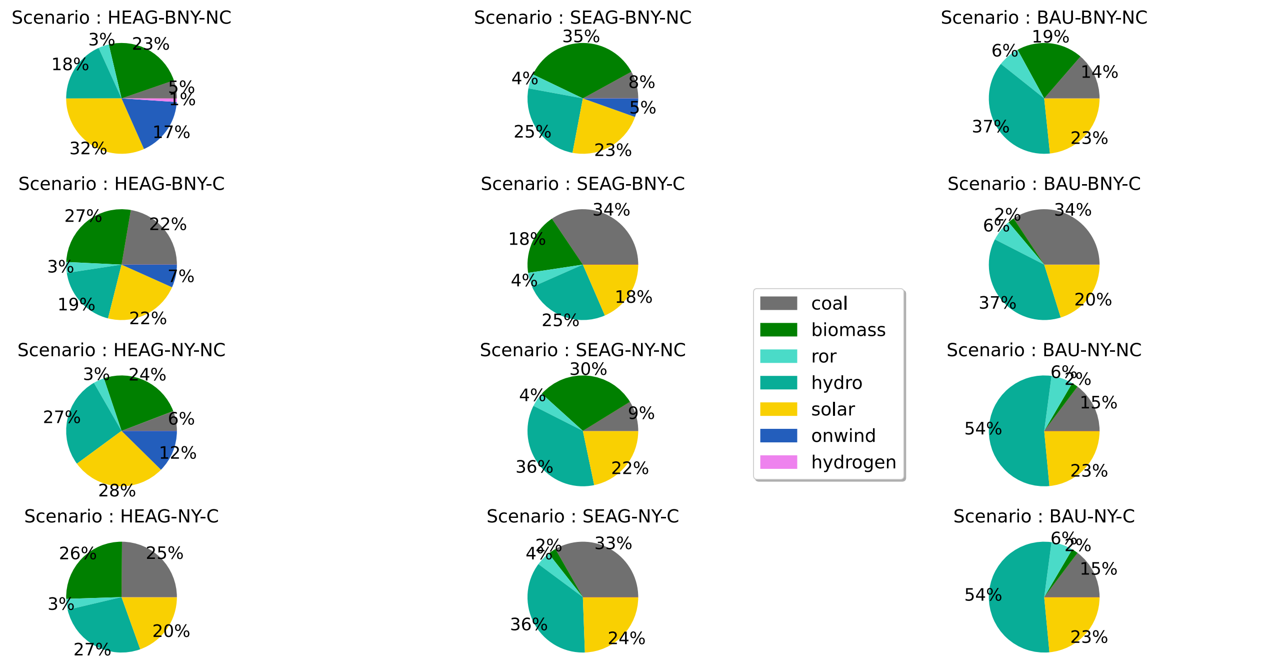


Fig. 14. Contribution of different carriers to meeting future energy demand in Zambia. For comparison, total electricity consumption in HEAG scenarios is 38.4 TWh, SEAG is 28.8 TWh, BAU is 19.2 TWh.

high electricity demand and low water availability. To ensure water resilience and maintain hydropower's role in stabilizing the grid, policies should focus on water resource management and the development of more drought-resilient small-scale hydropower plants.

Investments in solar and wind energy are essential to reduce reliance on hydropower and fossil fuels. Solar, in particular, can meet peak demand during sunny hours and contribute excess energy for storage, while wind provides a complementary supply profile. Investments in renewable energy will have to be made by both public and private sectors to realize the huge expansion needs. As such, policies should facilitate the deployment of these technologies by streamlining permitting processes, offering tax incentives, and establishing feed-in tariffs or auction systems that attract investment.

The results showed that access to financing mechanisms is critical to implementing these transitions. Here, it is important to distinguish between different financing mechanisms. Firstly, the significant upfront investment costs for renewables, and secondly, the reliance on coal in case of scalability challenges for biomass. Zambia can leverage international climate funds and establish public-private partnerships to reduce investment risks. For renewables, access to green funds, such as the Green Climate Fund, would encourage financing for renewable energy projects in Zambia. For coal, access to funds will be more challenging. Therefore, clear long-term targets, coupled with transparent regulatory frameworks, can provide investor confidence and drive sustained funding for the energy sector.

This finding is consistent with other studies in the literature, where the potential of renewable energy in the African context represents a huge opportunity to attract foreign capital given the enhanced policy frameworks [74–76]. A recent study by Mhango 2024 identified several drivers and barriers affecting the deployment of renewable energy technologies in Zambia [77]. It showed that government commitment emerged as a key driver, with strategic policies and plans emphasizing diversification of the energy mix and reducing reliance on hydropower. The existence of abundant renewable resources, particularly solar, and international partnerships were also highlighted as critical enablers. However, barriers such as overreliance on hydropower, outdated infrastructure, financial constraints, and limited private sector involvement hinder progress. By aligning these policies, technologies, and investment frameworks, Zambia can achieve a low-carbon, resilient energy future that balances universal access, renewable energy integration, and energy security.

Mobilizing private capital for Zambia's energy transition—particularly in renewable energy and off-grid electrification—requires addressing a complex set of investment risks. These include currency volatility, off-taker risk (e.g., delayed payments from utilities), regulatory uncertainty, and limited access to long-term, affordable finance. Such risks elevate the weighted average cost of capital (WACC), making projects less bankable and slowing progress toward universal access [78–81].

To mitigate these risks, a layered de-risking strategy that combines financial instruments, institutional reforms, and regional integration can be adopted [82,83]. Key measures include:

1. Bankable power purchase agreements (PPAs) with sovereign guarantees or escrow-backed payment mechanisms to reduce off-taker risk [84].
2. Currency risk mitigation through local currency-denominated contracts or blended finance structures that include concessional capital from development finance institutions [85].
3. Political risk insurance and partial risk guarantees from multilateral agencies such as the Multilateral Investment Guarantee Agency, the African Development Bank (AfDB), and the African Trade Insurance Agency [86].
4. Green bonds and climate-aligned finance, which can mobilize institutional investors seeking compliant portfolios [87,88].

Peer countries offer valuable lessons. Kenya's Scaling Solar and Feed-in Tariff programs have successfully attracted private investment by bundling PPAs with World Bank guarantees [78]. Morocco's Noor solar complex leveraged concessional finance and risk-sharing from the Clean Technology Fund and AfDB to reduce project risks and attract global investors [87]. South Africa's Renewable Energy Independent Power Producer Procurement Programme is widely regarded as a benchmark for transparent, competitive auctions that de-risk projects through standardized contracts and grid access guarantees [89].

Zambia can build on these models by strengthening the independence of its energy regulator, streamlining permitting, and enhancing tariff transparency. Participation in the Southern African Power Pool also offers opportunities to hedge market and supply risks through regional electricity trade. Embedding these measures into Zambia's energy strategy will be critical to unlocking the estimated $3–5 billion in investment needed to achieve universal access by 2030 [90].

6.4. What is the most cost-effective approach to integrating off-grid electrification solutions alongside grid expansion?

A hybrid approach that balances grid expansion with off-grid electrification solutions is the most cost-effective strategy for Zambia. A recent study on two rural areas in East Africa suggests that this hybrid strategy accompanied by proper dimensioning would improve mini-grids' financial profitability [91]. Moreover, usage of off-grid solutions in last-mile rural electrification programs should be tested in the local context and promoted by stakeholders and electrification authorities [92].

In urban and peri-urban areas, extending the national grid is economical due to existing infrastructure. However, in remote and sparsely populated regions, decentralized renewable energy solutions such as mini-grids and standalone solar home systems are more viable. Solar mini-grids in rural communities can provide reliable electricity without requiring costly transmission infrastructure. Additionally, government incentives, public-private partnerships, and community-based financing models can enhance affordability and adoption of off-grid systems. Smart grid integration and battery storage can further optimize the synergy between on-grid and off-grid solutions, ensuring cost-effective energy access nationwide.

6.5. How can the combination of grid and off-grid solutions enhance electricity access while ensuring long-term sustainability and affordability?

Integrating both grid and off-grid solutions optimizes electricity access by leveraging centralized and decentralized approaches. Grid expansion should prioritize high-demand and economically active regions, while off-grid systems should target remote areas where grid extension is financially impractical. Hybrid microgrids, which can be interconnected with the main grid when needed, provide flexibility and resilience. Energy-efficient technologies, smart meters, and demand-side management can further enhance sustainability by reducing overall consumption and improving efficiency.

Recent experience from energy transformation in other developing countries shows that universal access to energy cannot be exclusively achieved via direct electrification [93–97]. Government policies supporting net metering and feed-in tariffs can incentivize distributed renewable energy generation, making electricity more affordable and sustainable in the long term. This combined approach ensures universal electricity access while balancing cost, reliability, and environmental sustainability.

7. Conclusion

This research confirms that increasing electricity access and building a reliable electricity system in Zambia is both urgent and achievable through strategic diversification and resilience-building. With Zambia's current over-reliance on hydropower (84 % of its electricity mix), the study highlights the vulnerability of the system to climate variability, as seen in scenarios like BAU-BNY. Diversification, particularly through solar and wind energy, is essential for ensuring energy security. The analysis emphasizes the need for investments in grid flexibility and storage to address the challenges posed by renewable energy variability and growing electricity demand.

While coal pathways provide immediate dispatchable capacity, they are unsustainable in the long run due to high emissions and costs. Renewable-dominant scenarios offer optimal sustainability and system reliability, despite higher initial investments. Policy frameworks that include feed-in tariffs, tax incentives, and public-private partnerships are crucial to attract private sector involvement and accelerate the deployment of renewable technologies.

Overall, the analysis shows that achieving universal electrification in Zambia by 2030 is not only a matter of expanding infrastructure but also of optimizing cost-effectiveness and technology choices. The results showed that diversifying the energy system is key to enabling a robust and climate-resilient energy system. Moreover, diversifying electricity access strategies will be key to achieving higher national access goals with limited financial resources. This research establishes a compelling argument for Zambia to increase electricity access and build a reliable power system by 2030. Prioritizing renewable energy integration, modernizing the grid, and leveraging innovative financing mechanisms will mitigate climate risks, drive economic growth, and ensure a sustainable energy future.

While this study focuses on Zambia, its findings carry important implications for other developing countries facing similar challenges. According to the IEA, over 90 % of the global population currently has access to electricity, yet around 645 million people are still projected to remain without access by 2030, of whom 85 % will be in sub-Saharan Africa [98]. This underscores the urgent need for targeted, inclusive strategies. The analysis highlights that achieving universal electricity access requires strong support for decentralized solutions and a diversified approach to electrification. Integrating off-grid and mini-grid systems into national energy strategies can deliver more cost-effective access, especially in rural and remote areas. Moreover, diversifying the energy mix and investing in flexible, resilient technologies are essential not only for expanding access but also for enhancing supply reliability and driving sustainable economic growth.

This study offers a robust foundation for understanding Zambia's energy transition, however, several areas warrant further investigation. Future research should take a more integrated and forward-looking approach to support Zambia's evolving energy landscape. Key areas of focus include expanding scenario modelling to incorporate regional power interconnections, particularly Zambia's role within the Southern African Power Pool and possible linkages with the Eastern African Power Pool. Such interconnections could enhance energy security, enable electricity trade, and support grid balancing under variable renewable energy conditions. Second, integrating transport electrification and clean cooking solutions into the modelling framework would provide a more holistic view of Zambia's energy system and its alignment with broader development goals. Finally, strengthening the coupling between power system models and macroeconomic models (e.g., computable general equilibrium or input-output frameworks) will also be vital to assess the broader socio-economic impacts of different energy transition pathways, including employment, income distribution, and industrial competitiveness. Together, these research directions will provide a stronger foundation for evidence-based policymaking and a more resilient and equitable energy future for Zambia and other developing nations.

CRediT authorship contribution statement

Anas Abuzayed: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Albert Chitandula:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kumbuso Joshua Nyoni:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.enpol.2025.114786.

Appendix A. Regional granularity of the model

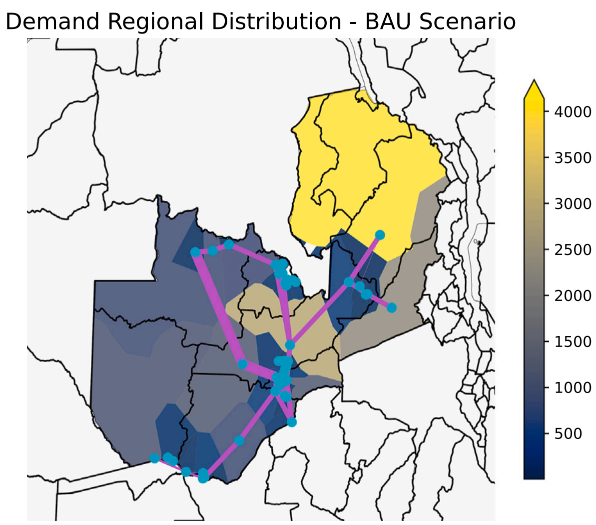

Fig. A.1. Regional electricity demand distribution under the Business-as-Usual (BAU) scenario shown over 50 nodes. The nodes (shown in blue) are extracted from the high voltage infrastructure in PyPSA-Earth using OpenStreetMap, alongside the available grid topology and transmission lines (shown in orchid) [12,37]. The spatial distribution of demand (shown in the colored map) is based on Voronoi cells around GIS-located substations, where each area is defined as the region closest to a given substation centroid [60]. Using GDP and population data, which serve as proxies for economic activity and residential load, the spatial granularity of demand enables thorough analysis of centralized versus decentralized electrification strategies and reflects the spatial modeling approach described in Section 3. The distribution pattern shown for BAU is representative and remains consistent across the SEAG and HEAG scenarios. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Data availability

The study is implemented on an open-source model with open-access data. Links are provided for the code and the data used.

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